

Week 05: Type classes and instances

Functional Programming in Lean

Learning goals

- ▶ class declarations
- ▶ instance declarations
- ▶ Instance synthesis

Core concepts

- ▶ `Label  $\alpha$` : overloaded operation
- ▶ `[Label  $\alpha$ ]`: class parameter
- ▶ `instance`: implementation for a type

Lean example

```
class Label ( $\alpha$  : Type) where  
  label :  $\alpha \rightarrow \text{String}$ 
```

```
def describe [Label  $\alpha$ ] (x :  $\alpha$ ) : String :=  
  Label.label x
```

Lean details

- ▶ Type class constraint at describe
- ▶ Instance synthesis at each call site
- ▶ Missing-instance diagnostic

Live coding

- ▶ `Label TrafficLight instance`
- ▶ `Label Nat instance`
- ▶ `#check describe`

Check

Instance used by describe `TrafficLight.green`

Exercise

- ▶ Instance for a second type
- ▶ Missing-instance diagnosis